Yebon Byun

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Education

University of California, Berkeley Berkeley, CA
Bachelor of Arts in Data Science May 2026

- **Concentration:** Applied Mathematics and Modeling
- **Coursework:** Machine Learning, Optimization, Data Science and Object Oriented Programming, Data Structure, Software Engineering, Probability and Statistics for Data Science, Advanced Linear Algebra
- **Leadership:** President, KSEA (Korean-American STEM Club) — Led 50+ members to organize seminars and group projects

Experience

Mediazen R&D Labs - Seoul, South Korea Sep 2024 - Dec 2025
Software Engineer, Machine Learning

- Delivered 98% F1-score in sensitive data de-identification by serving 1K+ daily privacy queries in an AI Contact Center through engineering and deploying a BiLSTM-CRF model in PyTorch with 500K+ utterances, and hyperparameter tuning.
- Enhanced BLEU score from 24% (27.5 → 34.2) in a bilingual English–Korean AI note-taking service by fine-tuning Meta's 418M-parameter LLM (M2M-100) model with HuggingFace Transformers and RAG-based context modeling.

NAVER Z - Seongnam, South Korea Dec 2023 – Mar 2024
Software Engineer, Machine Learning Intern

- Streamlined dataset annotation time by 38% (8h → 5h) for youth online safety by deploying a keyword-based detection system on 300K+ grooming-related messages, achieving 92% accuracy and enabling safety-focused ML dataset creation.

Climformatics - Richmond, California Sep 2022 – May 2023
Software Engineer, Machine Learning Intern

- Accelerated ETL pipeline runtime from 2d to 8h to support climate modeling for wine production at a Napa Valley winery by optimizing Python/Db2 pipelines, removing 15% duplicates and 10% missing values across 500K+ records.

National Security Innovation Network - Berkeley, California Jan 2022 - May 2022
Software Engineer, Machine Learning Intern

- Optimized ML training and inference runtime by 30% (10h → 7h) in Department of Defense 3D simulation research by parallelizing GPU workloads with CUDA on UC Berkeley Savio HPC clusters.
- Expanded 3D scene synthesis from single-room to house-level for defense simulation use cases by extending the ATISS data pipeline in Python to parse 18K+ houses and 86K+ rooms from the 3D-FRONT dataset.

Projects

Hackathon

- Enhanced YouTube's collaborative playlist feature by building a shared playlist creation and video playback service using the YouTube Data API, Flask, AJAX, and MongoDB; deployed on AWS EC2.

Gitlet

- Developed a Git-like version control system for efficient local source code management by implementing commit, checkout, branch, and merge operations in Java with serializable objects to persist commit histories.

Hybrid Mobile App Development

- Built a production-ready iOS/Android community app to 1K+ users supporting real-time communication and engagement by building a React Native frontend with Flask/MongoDB backend on AWS cloud infrastructure.

Technical Skills

Programming: Python, Java, C++, JavaScript, R, SQL, Bash, MATLAB
ML / Data: PyTorch, TensorFlow, Transformers, ONNX, Scikit-learn, Pandas, NumPy, Ray Tune, Gradio, MeCab, NLTK, CUDA, Spark
Systems / Cloud: Linux, AWS EC2, Docker, MongoDB, Git
Frameworks / Tools: Flask, FastAPI, Node.js, React Native, Jira, Confluence